

Ericsson Mobile Network Operations

May 2025

Contents

1	Introduction	3
2	Ericsson's Role in Mobile Network Operations	3
2.1	Historical Context	3
2.2	Key Offerings	3
3	Network Architecture	4
3.1	Radio Access Network (RAN)	4
3.2	Core Network	4
3.3	Transport Network	4
4	Network Management with Ericsson Network Manager	4
4.1	Key Features	4
4.2	Use Cases	5
5	Automation and AI in Network Operations	5
5.1	Machine Intelligence	5
5.2	Automation Frameworks	5
5.3	Energy Efficiency	5
6	Security in Mobile Networks	5
6.1	Key Security Features	5
6.2	Mission-Critical Networks	6
7	5G Evolution and Use Cases	6
7.1	Key Use Cases	6
7.2	Case Study: Sporting Event	6
7.3	Case Study: Industrial IoT	6
8	Energy Management and Sustainability	6
8.1	Techniques	6
8.2	Sustainability Metrics	7
9	Challenges in Mobile Network Operations	7
9.1	Network Complexity	7
9.2	Traffic Growth	7
10	Future Directions	7
10.1	6G Vision	7

10.2 Open RAN	7
10.3 Network APIs	7
11 Technical Deep Dive: 5G RAN	7
11.1 Massive MIMO	8
11.2 Dynamic Spectrum Sharing	8
11.3 RAN Slicing	8
12 Network Slicing Implementation	8
12.1 Configuration Steps	8
12.2 Challenges in Slicing	8
13 Traffic Analysis and Optimization	8
13.1 Tools and Techniques	8
14 Training and Skill Development	9
15 Case Studies	9
15.1 Smart City Deployment	9
15.2 Rural Connectivity	9
15.3 Automotive V2X	9
16 Advanced Network Analytics	9
16.1 Key Metrics	9
17 Global Deployment Examples	9
18 Regulatory and Compliance Considerations	10
18.1 Spectrum Management	10
19 Glossary	10
20 References	10

1 Introduction

This knowledge base provides an in-depth exploration of Ericsson Mobile Network Operations, detailing the technologies, strategies, and solutions that enable efficient, scalable, and secure mobile networks. As a global leader in telecommunications, Ericsson drives innovation in mobile network operations, supporting the transition from 4G to 5G and preparing for 6G. This document covers network architecture, management systems, automation, security, sustainability, and real-world applications, offering a comprehensive resource for operators, engineers, and stakeholders.

Mobile network operations involve planning, deploying, optimizing, and maintaining network infrastructure to deliver seamless connectivity. Ericsson's solutions address the growing demands for high-speed data, ultra-low latency, and energy efficiency, particularly in the 5G era. With contributions to 3GPP standards and a presence in over 180 countries, Ericsson supports more than 2 billion subscribers worldwide. This knowledge base is structured to provide technical depth, practical guidance, and insights into future trends.

2 Ericsson's Role in Mobile Network Operations

Ericsson is a pioneer in telecommunications, providing end-to-end solutions for radio access networks (RAN), core networks, transport networks, and managed services. Its portfolio enables operators to deliver enhanced mobile broadband (eMBB), massive machine-type communications (mMTC), and ultra-reliable low-latency communications (URLLC), critical for industries like healthcare, automotive, and manufacturing.

2.1 Historical Context

Founded in 1876, Ericsson has evolved from manufacturing telegraph equipment to leading 5G innovation. Its contributions to 2G, 3G, and 4G networks established the foundation for 5G advancements, including network slicing, massive MIMO, and cloud-native architectures. Ericsson's managed services and AI-driven solutions address modern challenges like network complexity, scalability, and sustainability.

2.2 Key Offerings

Ericsson's mobile network operations portfolio includes:

- **Radio Access Network (RAN):** Modular hardware and software for 4G and 5G, including base stations and antennas.
- **Core Network:** Cloud-native 5G standalone (SA) solutions supporting network slicing and service-based architecture.
- **Transport Network:** High-capacity fronthaul and backhaul solutions, including microwave and optical systems.
- **Network Management:** Ericsson Network Manager (ENM) for real-time monitoring, automation, and analytics.
- **Managed Services:** Outsourcing for network planning, optimization, and maintenance, reducing operational costs.
- **OSS/BSS:** Operations and Business Support Systems for service orchestration and monetization.

3 Network Architecture

Ericsson's mobile network architecture is designed for flexibility, scalability, and resilience, supporting diverse 5G use cases. The architecture comprises three core components: RAN, core network, and transport network, each optimized for performance and efficiency.

3.1 Radio Access Network (RAN)

The RAN connects user devices to the core network via base stations. Ericsson's Radio System includes remote radio units (RRUs), baseband units (BBUs), and advanced antenna systems (AAS). Features like beamforming, massive MIMO, and dynamic spectrum sharing (DSS) enhance coverage, capacity, and spectrum efficiency. For example, Ericsson's 64T64R massive MIMO radios support high-density urban deployments, delivering up to 3x capacity improvements over 4G.

3.2 Core Network

Ericsson's 5G core network, built on cloud-native principles, supports network slicing, service-based architecture (SBA), and containerized network functions (CNFs). It enables dynamic resource allocation and low-latency services for applications like autonomous vehicles and industrial IoT. The dual-mode 5G Core supports both non-standalone (NSA) and standalone (SA) deployments, ensuring a smooth transition from 4G.

3.3 Transport Network

The transport network links RAN and core components, requiring high bandwidth and low latency. Ericsson's transport solutions, including Router 6000 series and MINI-LINK microwave systems, support evolved CPRI (eCPRI), IP/MPLS, and dense wavelength division multiplexing (DWDM). These solutions ensure scalability for 5G's increased data demands, with latency as low as 1 ms for fronthaul connections.

4 Network Management with Ericsson Network Manager

Ericsson Network Manager (ENM) is a unified platform for managing radio, transport, and core networks. It provides real-time visibility, automation, and analytics, reducing operational complexity and improving network reliability.

4.1 Key Features

- **Automation:** Policy-based auto-scaling and AI-driven incident management.
- **Scalability:** Supports networks with thousands of nodes, from small cells to macro sites.
- **Integration:** Open APIs for seamless integration with third-party OSS and BSS systems.
- **Analytics:** Real-time monitoring, anomaly detection, and predictive maintenance.
- **User Interface:** Intuitive dashboards for network performance visualization.

4.2 Use Cases

ENM enables network slicing for differentiated services, automated fault detection, and energy optimization. For instance, during a major European sporting event in 2024, ENM managed 5G network slicing to deliver seamless connectivity to 100,000+ users, ensuring low latency for live streaming and social media.

5 Automation and AI in Network Operations

Automation is essential for managing the complexity of 5G networks. Ericsson leverages AI, machine learning, and intent-based networking to enhance efficiency, reduce downtime, and optimize resources.

5.1 Machine Intelligence

Ericsson's AI solutions include visual object detection, knowledge extraction, and predictive analytics. For example, its Augmented Reality (AR) troubleshooting tool guides technicians in resolving base station issues, reducing maintenance time by up to 30%. Machine learning models predict traffic spikes, enabling proactive resource allocation.

5.2 Automation Frameworks

Ericsson's automation frameworks, such as Cognitive Network Operations, use closed-loop automation to handle tasks like fault resolution and capacity planning. These frameworks integrate with ENM, enabling zero-touch operations for routine tasks.

5.3 Energy Efficiency

Ericsson's energy-efficient solutions, like micro-sleeps in packet processing and adaptive power management, reduce consumption by up to 68% at low traffic loads. For example, Ericsson's Radio Deep Sleep feature powers down unused components during off-peak hours, supporting sustainability goals.

6 Security in Mobile Networks

Security is critical in 5G networks due to increased device connectivity and mission-critical applications. Ericsson's 5G security framework ensures resilience, privacy, and availability across all network layers.

6.1 Key Security Features

- **Network Slicing:** Isolates critical functions to prevent interference and attacks.
- **Authentication:** Supports 5G Authentication and Key Agreement (AKA) and Extensible Authentication Protocol (EAP).
- **Encryption:** Secures user and control plane data in centralized and distributed units.
- **Threat Detection:** AI-based anomaly detection for real-time threat mitigation.

6.2 Mission-Critical Networks

Ericsson's mission-critical solutions, such as push-to-talk over LTE and 5G, meet stringent requirements for availability and quality of service (QoS). These solutions support public safety, emergency services, and industrial applications like smart grids.

7 5G Evolution and Use Cases

5G transforms industries through high-speed connectivity, ultra-low latency, and massive device support. Ericsson's 5G platform enables standalone (SA) networks, network slicing, and edge computing.

7.1 Key Use Cases

- **Fixed Wireless Access (FWA):** Delivers broadband to underserved areas with quick ROI.
- **Cellular IoT:** Supports industrial automation, smart cities, and massive IoT deployments.
- **Immersive Technologies:** Enables AR/VR for gaming, training, and remote collaboration.
- **Automotive:** Supports vehicle-to-everything (V2X) communication for autonomous driving.

7.2 Case Study: Sporting Event

During a 2024 European sporting event, Ericsson deployed 5G networks with network slicing, achieving peak data rates of 1 Gbps and latency below 10 ms. This enabled real-time streaming and enhanced fan experiences, such as AR overlays for live matches.

7.3 Case Study: Industrial IoT

Ericsson partnered with a European manufacturer to deploy a private 5G network for factory automation. The network supported 10,000+ IoT devices, reducing latency to 5 ms and improving production efficiency by 25%.

8 Energy Management and Sustainability

Ericsson's sustainability initiatives reduce network energy consumption, aligning with global environmental goals. For example, Kiona's Edge AI, powered by Ericsson's connectivity, saved 17.3 million kWh of energy, avoiding 1 kTonne of CO2 emissions.

8.1 Techniques

- **Micro-Sleeps:** Reduces processor energy use in data centers and base stations.
- **Optimized Resource Allocation:** Enhances throughput ramp-up for better efficiency.
- **AI-Driven Optimization:** Predicts traffic patterns to minimize power usage.

8.2 Sustainability Metrics

Ericsson's 2024 sustainability report highlights a 35% reduction in network energy consumption since 2019, driven by advanced hardware and software optimizations.

9 Challenges in Mobile Network Operations

Operators face challenges like network complexity, rising traffic, and regulatory compliance. Ericsson addresses these through scalable architectures, automation, and open standards.

9.1 Network Complexity

5G networks involve thousands of nodes, diverse use cases, and multi-vendor environments. Ericsson's ENM and automation tools simplify management, reducing operational overhead.

9.2 Traffic Growth

Global mobile data traffic is projected to grow by 20% annually through 2030. Ericsson's dynamic spectrum sharing and massive MIMO solutions optimize spectrum usage to handle this growth.

10 Future Directions

Ericsson is preparing for 6G and Open RAN, focusing on AI-native networks, terahertz frequencies, and network disaggregation.

10.1 6G Vision

6G will enable holographic communications, AI-driven orchestration, and ultra-low power devices. Ericsson's research focuses on integrated sensing and communication (ISAC) and sustainable architectures.

10.2 Open RAN

Ericsson supports Open RAN through cloud-native RAN solutions, promoting interoperability and vendor diversity. Its Cloud RAN platform allows operators to deploy virtualized RAN functions on commodity hardware.

10.3 Network APIs

Ericsson's Network Exposure Function (NEF) provides APIs for developers to access 5G capabilities, enabling new services like real-time analytics and IoT orchestration.

11 Technical Deep Dive: 5G RAN

The 5G RAN is a cornerstone of Ericsson's mobile network operations. It supports advanced features like massive MIMO, beamforming, and dynamic spectrum sharing, optimized for high-density deployments.

11.1 Massive MIMO

Massive MIMO uses multiple antennas to improve capacity and coverage. Ericsson's 64T64R configurations support up to 400 MHz bandwidth, delivering multi-gigabit speeds in urban areas.

11.2 Dynamic Spectrum Sharing

Dynamic spectrum sharing enables 4G and 5G to coexist on the same spectrum, reducing deployment costs. Ericsson's DSS implementation ensures seamless user transitions, with minimal latency impact.

11.3 RAN Slicing

RAN slicing allocates radio resources to specific services, ensuring QoS for diverse use cases. Ericsson's solutions support up to 100 slices per site, with dynamic reconfiguration.

12 Network Slicing Implementation

Network slicing creates virtual networks tailored to specific use cases, such as eMBB, URLLC, and mMTC. Ericsson's end-to-end slicing solutions integrate RAN, core, and transport layers.

12.1 Configuration Steps

1. Define slice requirements (e.g., latency, bandwidth, reliability).
2. Configure RAN resources using ENM's slicing tools.
3. Allocate core network functions via cloud-native platforms.
4. Deploy and test slice with real-time analytics.
5. Monitor performance and adjust parameters dynamically.

12.2 Challenges in Slicing

Network slicing requires precise orchestration and resource isolation. Ericsson's automation tools mitigate challenges like slice interference and resource contention.

13 Traffic Analysis and Optimization

Ericsson's traffic analysis tools use AI to identify application-specific patterns, enabling operators to optimize resources. For example, optimizing throughput ramp-up reduced YouTube time-to-content by 62% in LTE networks.

13.1 Tools and Techniques

- **Traffic Classification:** Identifies high-priority applications like video streaming.
- **QoS Management:** Prioritizes traffic for low-latency services.
- **Predictive Analytics:** Forecasts demand to optimize resource allocation.

14 Training and Skill Development

Ericsson offers extensive training on mobile network operations, covering cellular technology, infrastructure maintenance, and 5G commissioning. Programs include:

- **Ericsson Academy:** Online and in-person courses on 5G and IoT.
- **Certification Programs:** Validate skills in RAN and core network management.
- **Hands-On Labs:** Simulate real-world network scenarios for technicians.

15 Case Studies

15.1 Smart City Deployment

Ericsson collaborated with a Middle Eastern city to deploy a 5G smart city network, supporting traffic management, public safety, and IoT services. The network handled 50,000+ devices with 99.999% reliability.

15.2 Rural Connectivity

Ericsson's FWA solutions provided broadband to rural African communities, achieving payback periods of 18 months and connecting 10,000 households.

15.3 Automotive V2X

Ericsson's 5G V2X solutions enabled vehicle-to-infrastructure communication in a European pilot, reducing accident response times by 40%.

16 Advanced Network Analytics

Ericsson's analytics platforms use big data and machine learning to predict network performance, optimize resources, and enhance user experience. Tools like Ericsson Expert Analytics provide insights into subscriber behavior and network health.

16.1 Key Metrics

- **Network Performance:** Measures latency, throughput, and packet loss.
- **Subscriber Experience:** Tracks QoE for applications like streaming and gaming.
- **Fault Prediction:** Identifies potential issues before they impact users.

17 Global Deployment Examples

Ericsson's solutions are deployed globally, from 5G urban networks in Europe to IoT networks in Asia. Notable deployments include:

- **North America:** 5G SA network for a major operator, supporting 1 million+ users.
- **Asia-Pacific:** Private 5G network for a manufacturing hub, enabling Industry 4.0.
- **Europe:** Open RAN pilot with multi-vendor interoperability.

18 Regulatory and Compliance Considerations

Mobile network operations must comply with local and international regulations, such as spectrum allocation and data privacy laws. Ericsson's solutions align with GDPR, FCC, and 3GPP standards, ensuring compliance while maximizing performance.

18.1 Spectrum Management

Ericsson works with regulators to optimize spectrum allocation, supporting licensed, unlicensed, and shared spectrum models for 5G.

19 Glossary

- **RAN:** Radio Access Network, connecting devices to the core network.
- **ENM:** Ericsson Network Manager, a unified management platform.
- **Network Slicing:** Virtual networks for specific use cases.
- **Massive MIMO:** Multi-antenna technology for enhanced capacity.
- **DSS:** Dynamic Spectrum Sharing for 4G/5G coexistence.

20 References

- Ericsson official website: www.ericsson.com
- 3GPP Release 15 and 16 specifications
- Ericsson Mobility Report, 2024
- Ericsson Sustainability Report, 2024